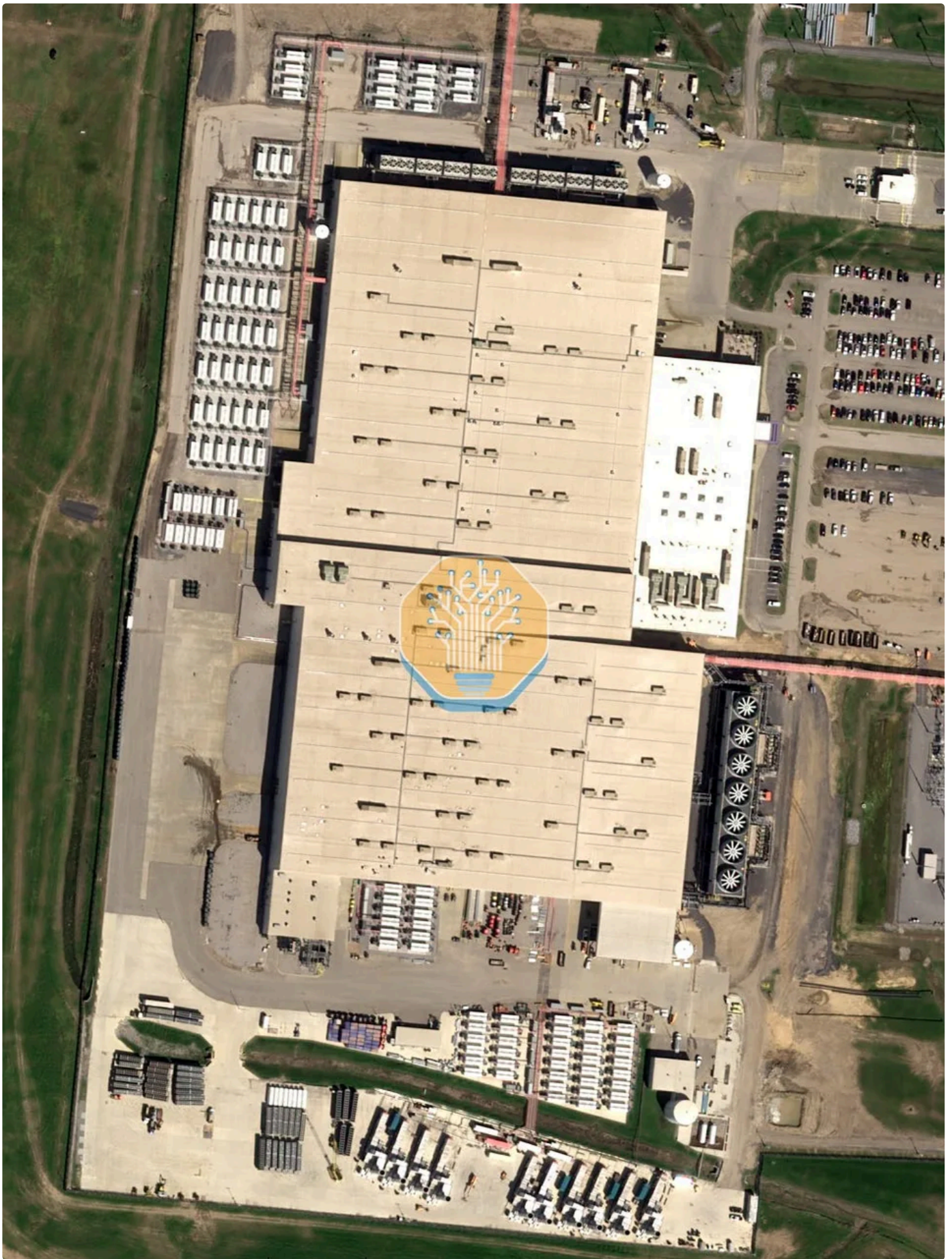




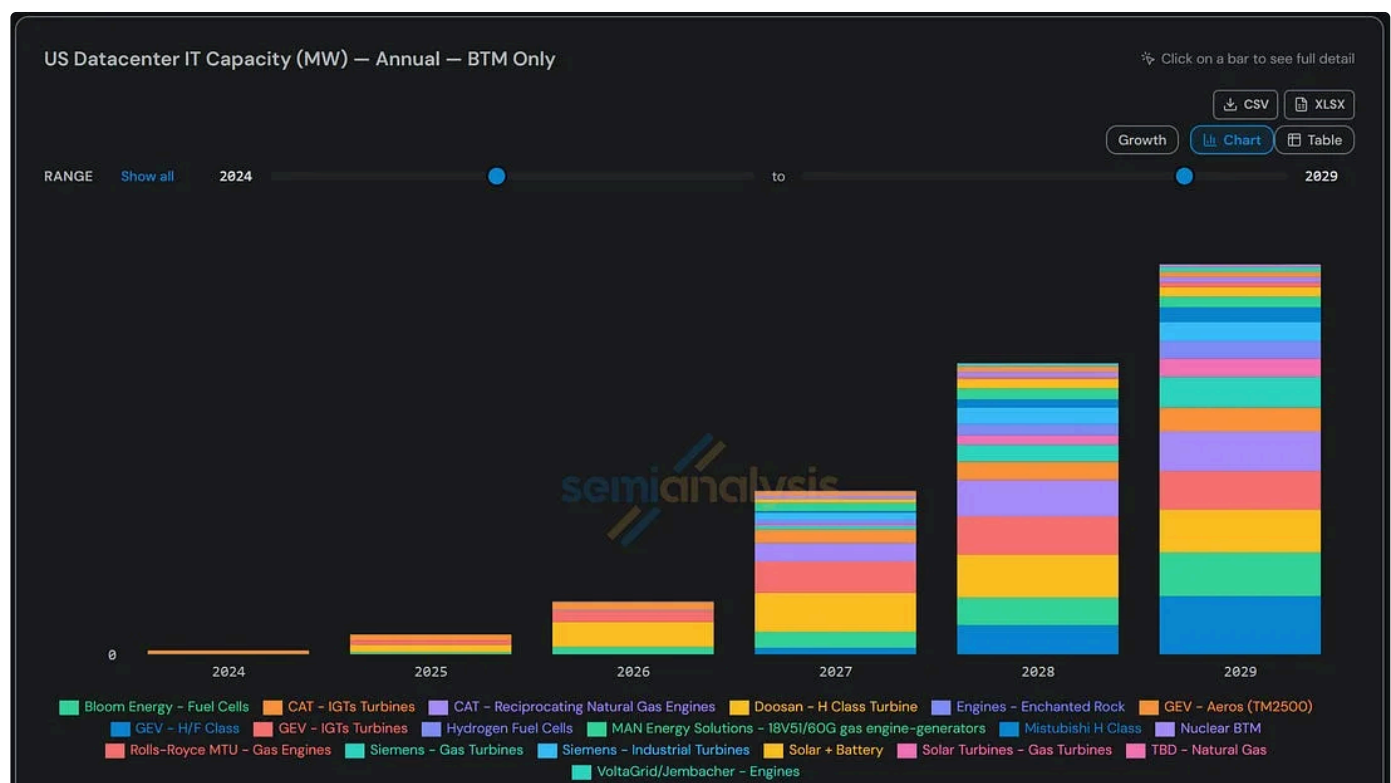
Source: SemiAnalysis Datacenter Industry Model; xAI's Colossus 1, retrofitting a warehouse

The physical playbook: energy and equipment

Grid connection lead times have stretched into 7–10 years in multiple metro areas, and equipment lead times for critical SKUs remain 1y+. The market has evolved adaptations on both axes.

On energy, the playbook has expanded from a single grid-connection path to multiple:

- **Pay directly into the grid.** Oracle’s deal with DTE to supply BESS systems to support their load, or Meta’s deal with Entergy where Meta pays for the full construction and operation of their dedicated power plant.
- **Powered Land.** In constrained markets like Europe and US metros, businesses are now *selling* or leasing grid-connected (or soon-to-be-connected) land outright. Lancium, Cloverleaf, and others are performing remarkably well in this market.
- **Behind-the-meter (BtM) generation.** One by one, hyperscalers and AI Labs and leading developers are giving up on the grid and going offsite. Read our onsite gas deep dive for more details.



Source: SemiAnalysis Datacenter Industry Model

On equipment, three adaptations have moved from edge cases to standard practice over the last 12 months:

- **Long-lead procurement in the scoping phase.** Hyperscalers now lock in transformers, MV/LV switchgear, and other long-lead SKUs well ahead of vertical construction. Our Industrials Model stays on top of the rising bottlenecks, and after Computex Taipei, we’ve flagged a specific equipment for cooling that likely becomes the next big bottleneck. Contact us for more information at sales@semianalysis.com.

- **Chinese OEMs in the supply chain.** Few will admit it publicly, but Chinese OEMs are quietly alleviating bottlenecks at the high end of the equipment stack. xAI, for example, likely uses Sieyuan transformers/breakers. Multiple US companies now broker for Chinese OEMs, and in some cases hyperscalers purchase directly. The trend has accelerated meaningfully over the last 12 months.



Source: Sieyuan

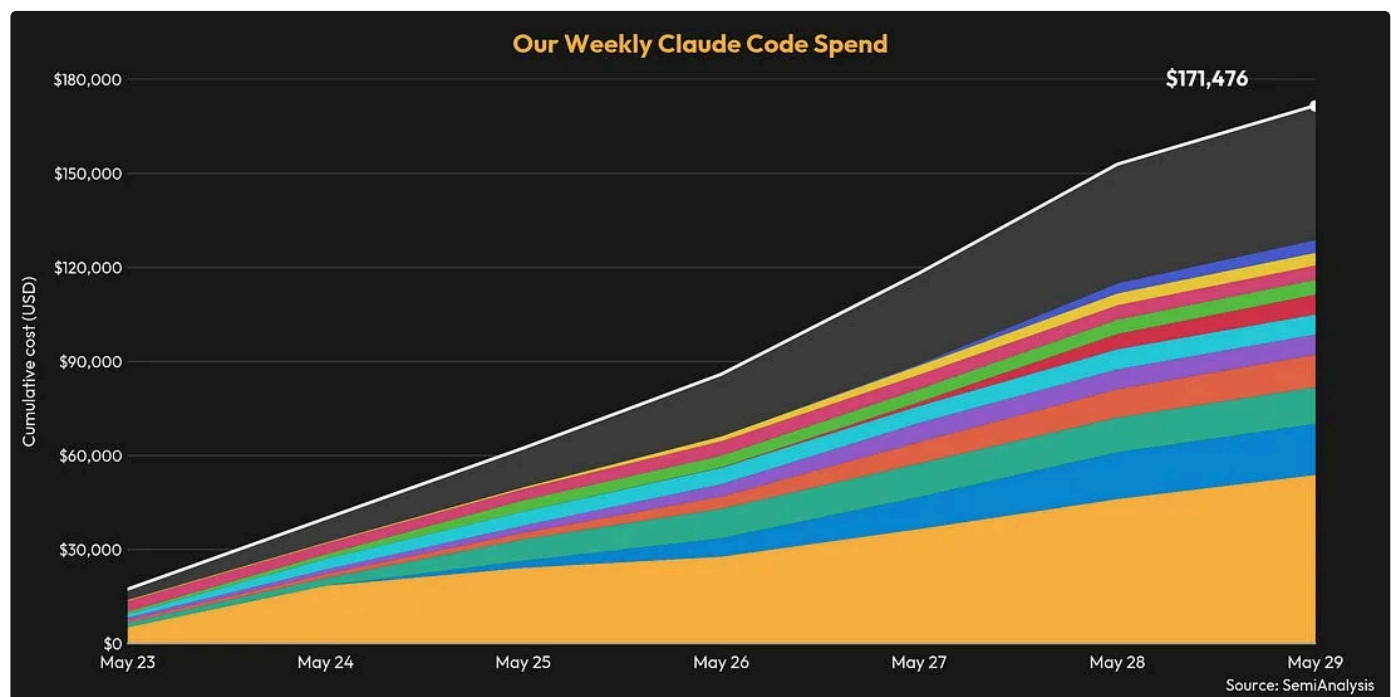
- **Modular and prefabricated builds.** Old news in the Chinese markets, *new* to American AI datacenters. Prefab condenses supply-chain complexity into a single product and dramatically reduces the skilled-labor intensity of onsite assembly, and the same hyperscalers paying inflated wages and onsite housing in Abilene are increasingly the ones leaning on prefab to sidestep that crunch on the next site. We have a dedicated deep dive forthcoming; for now, the takeaway is a meaningful pressure-release valve on equipment and labor constraints simultaneously.

The political playbook secures the first *right* to build, and the physical playbook ensures the build progresses accordingly.

What Do We Do Differently?

With all the above challenges and the varieties of workarounds, how are we able to determine veritable delays? Some projects may die on local opposition, some might pass and see the approving council members ousted. Labor is a challenge, or you can import from across the country. Even behind-the-meter sites are still a novel solution.

Our research agents monitor thousands of municipal, county, state, and utility portals — most poorly structured, many actively resistant to scraping, some still publishing in scanned PDFs and handwritten exhibits. We parse them in parallel and reconcile every signal against live satellite imagery, which is how we generate forward-looking delay calls across the full US pipeline rather than just the highest-profile projects.



Where we go beyond what's public

- Tasking custom satellite captures on specific sites, on specific days for faster cadence than commercial imagery providers default to.
- Scraping social media presence of NIMBY coalitions, construction firms, and local community pages for sentiment shifts before they reach local press.
- On-the-ground site visits and direct outreach to regulators for documents that were never digitized.

The full stack is not disclosed publicly.

The Datacenter Model methodology is coupled with the Industrials and Energy teams tracking the power and equipment/labor markets respectively. When every angle is tracked in parallel, it becomes straightforward to identify which projects carry real cancellation or delay risk, and which apparent "delays" are simply the market culling speculation that was never going to deliver in 2026. Behind the

paywall, we discuss another veritable delay coming up, what we are watching, and the players set to benefit.

The Equipment Supplier Narrative Is Wrong Too

We said at the top we would come back to the equipment suppliers. The fear priced into names like Vertiv is that the wave of datacenter delays and cancellations will start eating into record orders and backlog. However, our Datacenter Model capacity forecast shows quarter by quarter delivery, and it's clear to us that delivered MWs are accelerating.

Equipment lead times remain long. Reinhausen tap-changer bushings, which are a single subcomponent inside a transformer but gate delivery of the entire large HV stack, are now quoted at three to five years. Things have stabilized somewhat over the past year, though most critical SKUs are still running over a year, and several of the major manufacturers, including GE Vernova, Hitachi Energy, and Mitsubishi Electric, are booked out three to four years on their main equipment lines.

Capacity relief is slow to arrive, since production expansion can take two to three years; Hitachi Energy's new South Boston plant, announced in September 2025, is not expected to be operational until 2028. So in the meantime, the way capacity actually gets allocated is through prepayments. Buyers wire money up front to hold a slot in the queue, which boxes out players with less capital. For behind-the-meter systems, we have heard prepayments in the 10 to 15% range are now standard practice for securing a queue position. AI demand has lifted the pure-play datacenter electricals, Vertiv and Schneider among them, to margins above 20% on the strength of that same imbalance.

Now run the cancellation fear against that backdrop. The projects getting canceled sit in the early stage layer that never placed equipment orders in the first place; a speculative announcement dying in a county commission hearing removes zero orders from anyone's books. The projects that do sit in OEM backlogs have prepaid for queue position, locked in long-lead SKUs during scoping, and in many cases already broken ground. And in the rare case a real order does fall away, a queue running three to four years deep means the slot is reallocated to the next buyer in line rather than vanishing.

Our Industrials Model tracks more than 550 datacenter equipment suppliers across 75 equipment subcategories, mapped to over 6,000 facilities globally, and maps every major supplier's exposure project by project, which is how we separate backlog at risk from backlog that is locked.